C++ Question Bank

63 multiple-choice questions (Core 20, Easy 8, Medium 18, Hard 17) and 8 coding questions. Answer key on the last pages.

MCQs - Core (20)

1. Which of these is NOT one of the four pillars of object-oriented programming?

A) Encapsulation
B) Inheritance
C) Compilation
D) Polymorphism

2. What is the difference between a struct and a class in C++?

A) A struct cannot have member functions
B) Members and base classes are public by default in a struct and private by default in a class
C) A struct cannot be inherited
D) There is no difference at all

3. What is the time complexity of accessing v[i] in a std::vector?

A) O(1)
B) O(log n)
C) O(n)
D) O(n log n)

4. Which STL container gives average O(1) lookup by key (without ordering the keys)?

A) std::map
B) std::unordered_map
C) std::list
D) std::vector

5. What is the output?

```
vector<int> v = {4, 1, 3};
sort(v.begin(), v.end());
for (int x : v) cout << x;
```

A) 413
B) 134
C) 431
D) 314

6. What is the output?

```
queue<int> q;
q.push(1); q.push(2); q.push(3);
q.pop();
cout << q.front() << q.back();
```

A) 23
B) 12
C) 32
D) 13

7. Which is the correct way to release an array allocated with int *a = new int[10]; ?

A) delete a;
B) delete[] a;
C) free(a);
D) No release is needed

8. What is encapsulation?

A) Deriving a class from another
B) Bundling data with the methods that operate on it and restricting direct access to the data
C) Having many functions with the same name
D) Allocating objects on the heap

9. What is the output?

```
string s = "coding";
cout << s.substr(1, 3) << s.length();
```

A) odi6
B) cod6
C) odin6
D) odi5

10. Which feature lets a derived class provide its own implementation of a base class virtual function?

A) Overloading
B) Overriding
C) Templates
D) Friend functions

11. What is the output?

```
int a[] = {3, 7, 2, 9};
int mx = a[0];
for (int x : a) if (x > mx) mx = x;
cout << mx;
```

A) 3
B) 7
C) 9
D) 2

12. In a binary search tree, where is a value smaller than the root stored?

A) In the right subtree
B) In the left subtree
C) At a leaf only
D) Anywhere

13. What does the this pointer refer to inside a non-static member function?

A) The class itself
B) The object on which the function was called
C) The base class
D) The last created object

14. Which container is best when you need fast insertion and removal at both the front and the back?

A) std::vector
B) std::deque
C) std::set
D) std::array

15. What is the output?

```
int x = 5;
int *p = &x;
*p = *p * 2;
cout << x;
```

A) 5
B) 10
C) Address of x
D) 25

16. What is a pure virtual function?

A) A function that cannot be overridden
B) A virtual function declared with = 0 that makes the class abstract
C) A static function
D) An inline function with no body

17. What is the output?

```
map<char, int> freq;
for (char c : string("banana")) freq[c]++;
cout << freq['a'] << freq['n'] << freq.size();
```

A) 323
B) 322
C) 213
D) 331

18. Function overloading is resolved by:

A) The return type only
B) The number and types of the parameters
C) The function body
D) The name of the parameters

19. What is the worst-case time complexity of searching for a value in an unsorted std::vector?

A) O(1)
B) O(log n)
C) O(n)
D) O(n^2)

20. What is the output?

```
class Box {
    int w;
public:
    Box(int x) : w(x) {}
    int area() { return w * w; }
};
...
Box b(4);
cout << b.area();
```

A) 4
B) 8
C) 16
D) Compilation error: w is private

MCQs - Easy (8)

21. Which C++ feature allows a function to have the same name but different parameter lists?

A) Function overriding
B) Function overloading
C) Templates
D) Inline functions

22. What is the default access specifier for members of a class in C++?

A) public
B) protected
C) private
D) friend

23. What is the output?

```
int x = 10;
int &r = x;
r = 20;
std::cout << x;
```

A) 10
B) 20
C) Garbage value
D) Compilation error

24. Which STL container stores unique elements in sorted order?

A) std::vector
B) std::list
C) std::set
D) std::deque

25. Which operator cannot be overloaded in C++?

A) +
B) []
C) ::
D) ==

26. What does std::vector<int> v(5, 2); create?

A) A vector with elements 5 and 2
B) A vector of 5 elements, each equal to 2
C) A vector of 2 elements, each equal to 5
D) A compilation error

27. What is the output?

```
class A { public: void show(){ std::cout << "A"; } };
class B : public A { public: void show(){ std::cout << "B"; } };
A *p = new B(); p->show();
```

A) A
B) B
C) AB
D) Compilation error

28. What does const int *p mean?

A) p is a constant pointer to int
B) p is a pointer to a constant int (value cannot be changed through p)
C) Both pointer and value are constant
D) Invalid declaration

MCQs - Medium (18)

29. What is the output?

```
struct A { A() { cout << "A"; } ~A() { cout << "a"; } };
struct B : A { B() { cout << "B"; } ~B() { cout << "b"; } };
int main() { B obj; }
```

A) ABab
B) ABba
C) BAba
D) BAab

30. What is the output?

```
class C { public: static int n; C() { n++; } };
int C::n = 0;
int main() { C a, b[3]; cout << C::n; }
```

A) 1
B) 3
C) 4
D) 0

31. What is the output?

```
void f(int x)  { x = 10; }
void g(int &x) { x = 20; }
int main() { int a = 1; f(a); cout << a; g(a); cout << a; }
```

A) 1020
B) 120
C) 11
D) 1010

32. What is the output?

```
vector<int> v = {1, 2, 3};
v.push_back(4);
v.pop_back();
v.insert(v.begin(), 0);
cout << v.size() << v[0];
```

A) 30
B) 40
C) 41
D) 50

33. What is the output?

```
map<string, int> m;
m["b"] = 2; m["a"] = 1; m["c"] = 3;
for (auto &p : m) cout << p.first;
```

A) bac
B) abc
C) cba
D) Unspecified order

34. What is the output?

```
int f(int a, int b = 2, int c = 3) { return a + b + c; }
...
cout << f(1) << " " << f(1, 1);
```

A) 6 5
B) 6 6
C) 1 2
D) Compilation error

35. What is the output?

```
vector<int> v{1, 2, 3};
for (auto x : v)  x *= 2;
cout << v[0];
for (auto &x : v) x *= 2;
cout << v[0];
```

A) 22
B) 12
C) 11
D) 24

36. What is the output?

```
string s = "hello";
s += " world";
cout << s.find("o") << " " << s.size();
```

A) 4 11
B) 5 11
C) 4 10
D) 7 11

37. What is the output?

```
try { throw 5; }
catch (double) { cout << "d"; }
catch (int)    { cout << "i"; }
catch (...)    { cout << "all"; }
```

A) d
B) i
C) all
D) iall

38. What is the output?

```
int x = 10;
int main() { int x = 20; cout << x << ::x; }
```

A) 2020
B) 1010
C) 2010
D) 1020

39. What is the output?

```
set<int> s = {3, 1, 3, 2, 1};
cout << s.size() << *s.begin();
```

A) 51
B) 31
C) 33
D) 53

40. What is the output?

```
priority_queue<int> pq;
pq.push(3); pq.push(1); pq.push(4);
pq.pop();
cout << pq.top();
```

A) 1
B) 3
C) 4
D) Compilation error

41. What is the output?

```
class E {};
cout << sizeof(E);
```

A) 0
B) 1
C) 4
D) 8

42. What happens?

```
class A {
  int v = 0;
public:
  void set(int x) { v = x; }
};
int main() { const A a; a.set(5); }
```

A) It compiles and a.v becomes 5
B) Compilation error: a non-const member function cannot be called on a const object
C) Runtime error
D) It compiles, but the assignment is ignored

43. What is the output?

```
struct Shape { virtual void draw() = 0; };
int main() { Shape s; }
```

A) Nothing is printed
B) Compilation error: cannot declare a variable of abstract type
C) Runtime error: pure virtual call
D) Linker error

44. Why is virtual inheritance used, e.g. class B : virtual public A ?

A) To make all methods of A virtual
B) To make sure only one shared A sub-object exists in a diamond hierarchy
C) To prevent A from being inherited again
D) To allow A's private members to be accessed

45. What is the output?

```
void f(int)    { cout << "int "; }
void f(double) { cout << "double "; }
...
f('a');
f(2.5f);
```

A) int double
B) double double
C) int int
D) Compilation error (ambiguous)

46. What is the output?

```
stack<int> s;
s.push(1); s.push(2); s.push(3);
s.pop();
cout << s.top() << s.size();
```

A) 22
B) 32
C) 12
D) 23

MCQs - Hard (17)

47. What is the output?

```
struct A {
  A() {}
  A(const A &) { cout << "C"; }
};
void f(A a) {}
int main() { A x; f(x); A y = x; A z; z = x; }
```

A) CCC
B) CC
C) C
D) CCCC

48. What is the output?

```
struct B { virtual ~B() { cout << "~B"; } };
struct D : B { ~D() { cout << "~D"; } };
int main() { B *p = new D; delete p; }
```

A) ~B
B) ~D
C) ~D~B
D) ~B~D

49. What is the output?

```
struct B { B() { f(); } virtual void f() { cout << "B"; } };
struct D : B { void f() override { cout << "D"; } };
int main() { D d; }
```

A) B
B) D
C) BD
D) Undefined behaviour

50. What is the output?

```
struct Base    { virtual void print() { cout << "Base"; } };
struct Derived : Base { void print() override { cout << "Derived"; } };
void show(Base b) { b.print(); }
int main() { Derived d; show(d); }
```

A) Base
B) Derived
C) BaseDerived
D) Compilation error

51. What is the output?

```
int x = 1;
auto f = [x]  { return x; };
auto g = [&x] { return x; };
x = 5;
cout << f() << g();
```

A) 11
B) 55
C) 15
D) 51

52. What happens?

```
unique_ptr<int> a = make_unique<int>(5);
unique_ptr<int> b = a;
```

A) b and a both point to 5
B) Ownership is transferred to b, and a becomes null
C) Compilation error: unique_ptr cannot be copied
D) Runtime error (double delete)

53. What is the output?

```
auto a = make_shared<int>(5);
auto b = a;
{ auto c = a; }
cout << a.use_count();
```

A) 1
B) 2
C) 3
D) 0

54. What is the output?

```
struct B { virtual ~B() {} };
struct D : B {};
int main() {
  B *b = new B;
  D *d = dynamic_cast<D *>(b);
  cout << (d == nullptr);
}
```

A) 0
B) 1
C) Compilation error
D) Throws std::bad_cast

55. What is the output?

```
template <typename T> void f(T)  { cout << "T"; }
template <> void f<int>(int)      { cout << "int"; }
int main() { f(1); f(1.0); f('c'); }
```

A) intintint
B) intTT
C) TTT
D) intTint

56. What is the output?

```
struct E {};
struct F : E {};
int main() {
  try { throw F(); }
  catch (E &) { cout << "E"; }
  catch (F &) { cout << "F"; }
}
```

A) F
B) E
C) EF
D) Compilation error

57. What is the output?

```
struct B { void f(int)    { cout << "B"; } };
struct D : B { void f(double) { cout << "D"; } };
int main() { D d; d.f(1); }
```

A) B
B) D
C) Compilation error (ambiguous)
D) BD

58. What is the output?

```
map<int, int> m;
if (m[5] == 0) cout << m.size();
```

A) 0
B) 1
C) Nothing is printed
D) Throws std::out_of_range

59. Consider the code below. What is the value of a after S s; ?

```
struct S {
  int a, b;
  S() : b(5), a(b + 1) {}
};
```

A) 6
B) 1
C) Indeterminate - members are initialised in declaration order, so b is not yet set when a is initialised
D) Compilation error

60. What is the risk in this code?

```
vector<int> v = {1, 2, 3};
int &first = v[0];
v.push_back(4);
cout << first;
```

A) None - it prints 1
B) push_back may reallocate the storage, leaving `first` a dangling reference (undefined behaviour)
C) Compilation error
D) It always prints 4

61. What is the output?

```
int a = 5;
int &r = a;
int *p = &r;
*p = 7;
r++;
cout << a;
```

A) 5
B) 7
C) 8
D) 6

62. What is the output?

```
struct P { int v; P(int x) : v(x) {}
  P operator+(const P &o) const { return P(v * o.v); } };
int main() { P a(2), b(3), c(4); cout << (a + b + c).v; }
```

A) 9
B) 24
C) 10
D) Compilation error

63. What is the output?

```
int i = 10;
const int *p = &i;
int *const q = &i;
*q = 20;
cout << *p;
```

A) 10
B) 20
C) Compilation error at *q = 20
D) Undefined behaviour

Coding Questions (C++)

Every program reads from standard input and writes to standard output. Marks are awarded in proportion to hidden test cases passed.

C1. Centered Star Pyramid (Paper A)

Read the number of levels and print a centered star pyramid (example for levels = 5):

Example (levels = 5):

```
    *
   ***
  *****
 *******
*********
```

C2. First Standalone Character (Paper A)

Read a string of lowercase letters and find the first character that does not belong to any consecutive repeating group. A character belongs to a repeating group if the same character appears immediately before or after it. Print the first standalone character; if none exists print "No Character Found".

Sample input

```
aabbccdefggd
```

Sample output

```
d
```

C3. Missing Number in 1..N (Paper A)

The first line of input contains N. The second line contains N-1 distinct integers from 1 to N with exactly one number missing. Print the missing number.

Sample input

```
5
1 2 3 5
```

Sample output

```
4
```

C4. Prime Number Triangle (Paper A)

Read N and print a right-angled triangle of consecutive prime numbers with N rows, where row i contains i primes separated by single spaces.

Sample input

```
5
```

Sample output

```
2
3 5
7 11 13
17 19 23 29
31 37 41 43 47
```

C5. Pascal's Triangle (Paper B)

Write a program that reads an integer n from standard input and prints Pascal's Triangle with n rows. Print the numbers of each row separated by a single space, one row per line (left-aligned).

Sample input

```
4
```

Sample output

```
1
1 1
1 2 1
1 3 3 1
```

C6. Left-aligned Star Pyramid (Paper B)

Read n and print a left-aligned, increasing star pyramid with n rows where row i has 2*i-1 stars.

Sample input

```
5
```

Sample output

```
*
***
*****
*******
*********
```

C7. First Non-Repeating Character (Paper B)

Read a string and print the first character that appears exactly once in the whole string (left-to-right order). If every character repeats, print "None".

Sample input

```
aabbcdd
```

Sample output

```
c
```

C8. Top K Frequent Elements (Paper B)

The first line contains the integers of an array separated by spaces. The second line contains k. Print the k most frequent elements in descending order of frequency, separated by spaces (if two elements have the same frequency, print the smaller value first).

Sample input

```
1 1 1 2 2 3 4 4 4 4
2
```

Sample output

```
4 1
```

Paper A: C1-C4 Paper B: C5-C8

Answer Key

1. C	2. B	3. A	4. B	5. B	6. A	7. B	8. B
9. A	10. B	11. C	12. B	13. B	14. B	15. B	16. B
17. A	18. B	19. C	20. C	21. B	22. C	23. B	24. C
25. C	26. B	27. A	28. B	29. B	30. C	31. B	32. B
33. B	34. A	35. B	36. A	37. B	38. C	39. B	40. B
41. B	42. B	43. B	44. B	45. A	46. A	47. B	48. C
49. A	50. A	51. C	52. C	53. B	54. B	55. B	56. B
57. B	58. B	59. C	60. B	61. C	62. B	63. B	